

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

### CO2 ASSESSMENT TOOL 1 — Scenario-Based Requirement Analysis

|                       |                                                                                                                                                  |                      |                                                         |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------|
| <b>Course Code:</b>   | CSA1007                                                                                                                                          | <b>Course Title:</b> | Software Engineering                                    |
| <b>CO Assessed:</b>   | CO2                                                                                                                                              | <b>Assessment:</b>   | Assessment Tool 1 – Scenario-Based Requirement Analysis |
| <b>Weightage:</b>     | 20%                                                                                                                                              | <b>Total Marks:</b>  | 25                                                      |
| <b>CO2 Statement:</b> | <i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills.</i> |                      |                                                         |
| <b>Student Name:</b>  | V.Rakshana                                                                                                                                       | <b>Reg. No.:</b>     | 192425137                                               |
| <b>Date:</b>          | 30/07/2026                                                                                                                                       | <b>Duration:</b>     | 60 minutes                                              |

#### Code of Conduct

*I V.Rakshana(192425137), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: V.Rakshana

#### ***Problem Statement 24: AI-Based Fashion Supply Chain Optimization Platform***

## **1. Analyze System Requirements**

### **1.1 System Objectives**

The Smart Museum Artifact Preservation Platform aims to preserve valuable museum artifacts using Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, and predictive analytics. The system continuously monitors environmental conditions, predicts deterioration, schedules maintenance, maintains digital records, and assists museum administrators in making data-driven preservation decisions.

#### **Objectives**

- Preserve historical artifacts through continuous environmental monitoring.
- Predict artifact deterioration using Artificial Intelligence.
- Monitor temperature, humidity, lighting, vibration, and air quality using IoT sensors.
- Automate preventive maintenance scheduling.

- Maintain a centralized digital inventory of museum artifacts.

## **1.2 Functional Requirements**

### **Artifact Management**

- Register new artifacts.
- Update artifact information.
- Upload photographs and documentation.
- Categorize artifacts based on type and age.

## **1.3 Non-Functional Requirements**

### **Performance**

Dashboard should load within 3 seconds.

### **Scalability**

Support multiple museums and thousands of artifacts.

### **Reliability**

System availability of 99.5%.

## **2. Select Suitable Software Architecture**

### **Hybrid Microservices Architecture**

The Smart Museum Artifact Preservation Platform uses a Hybrid Microservices Architecture because each functional module operates independently while communicating through secure REST APIs and an API Gateway.

### **Major Microservices**

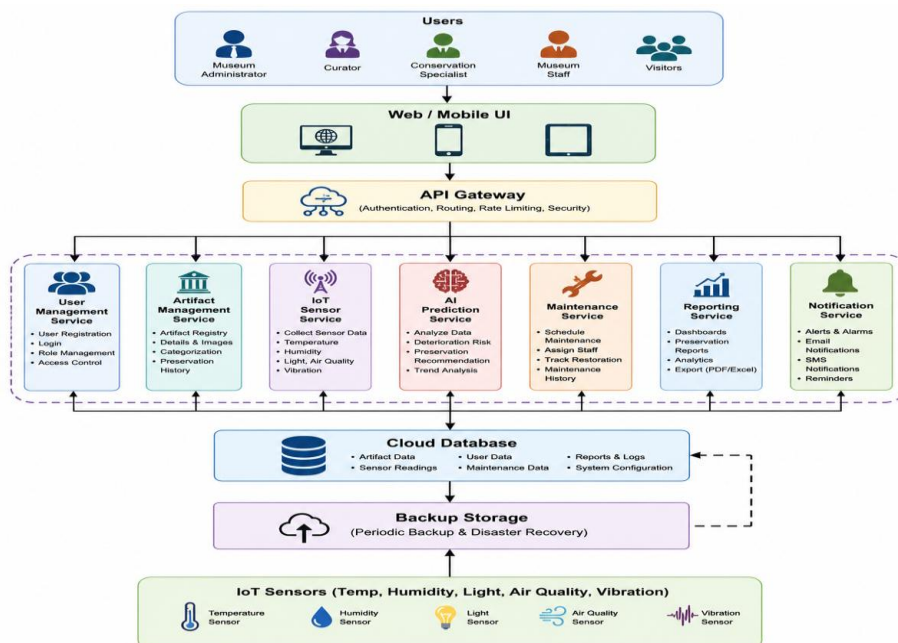
- User Management Service

- Artifact Management Service
- IoT Sensor Service

### Why Hybrid Microservices?

- Independent deployment
- Faster module development
- Cloud deployment support

## 3. Software Architecture Diagram



## 4. Components and Their Interactions

### User Interface

Provides web and mobile dashboards for museum administrators, curators, and conservation specialists.

### API Gateway

Routes requests to the appropriate microservices and provides authentication and security.

## **User Management Service**

- User Registration
- Role-Based Access Control
- Session Management

## **5. Workflow**

1. IoT sensors continuously monitor museum conditions.
2. Sensor data is sent to the IoT Service.
3. Data is stored in the cloud database.
4. AI analyzes environmental conditions.
5. AI predicts deterioration risk.
6. Alerts are generated if thresholds are exceeded.
7. Maintenance schedules are automatically created.
8. Administrators view dashboards and reports.

## **6. Future Enhancements**

- Digital Twin for museum visualization
- Blockchain-based artifact ownership records
- AI-based restoration recommendation engine
- Drone-based artifact monitoring
- Voice-enabled museum assistant

## **7. Conclusion**

The Smart Museum Artifact Preservation Platform uses a Hybrid Microservices Architecture integrated with AI, IoT, and cloud computing to provide an intelligent and scalable preservation solution. The architecture supports continuous monitoring, predictive analytics, automated maintenance, secure data management, and real-time reporting, making it well suited for modern museum environments while supporting future expansion and technological enhancements.